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Production versus Revenue Efficiency with Limited Tax Capacity: Theory and  
Evidence from Pakistan

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# 1 Big Picture

## 1.1 One-sentence summary

This note summarizes the paper, which studies whether governments with limited tax capacity should deliberately use production-inefficient taxes: combining a model of firm production and tax evasion with Pakistani corporate tax-return data, the paper exploits bunching at minimum-tax kinks to show that turnover taxation sharply reduces evasion and can raise corporate tax revenue substantially despite violating production efficiency.

## 1.2 Research question

The central question is whether a government should preserve **production efficiency** by taxing profits, or sacrifice some production efficiency by taxing a broader base such as turnover in order to improve **revenue efficiency** when tax evasion is severe.

In a standard Diamond–Mirrlees world with perfect enforcement, taxing profits dominates turnover taxation because a pure profit tax does not distort production decisions. But in low-capacity tax systems, profit taxes are easy to evade through overreported costs or hidden revenue. The paper asks whether this enforcement problem can justify taxing turnover, even though turnover taxation distorts firms’ production decisions and can generate cascading effects through supply chains.

## 1.3 Core answer

The paper’s answer is yes. When tax capacity is limited, a production-inefficient tax can be desirable if it greatly reduces evasion. In Pakistan, the authors estimate that switching from profit taxation to turnover taxation reduces evasion by up to **60–70 percent of corporate income**. In their calibrated model, a switch from profit taxation to turnover taxation can raise corporate tax revenue by about **74 percent** without lowering aggregate after-tax profits.

## 1.4 Key idea in simple terms

A profit tax asks the government to verify both revenues and costs. If firms can inflate costs, the profit base is easy to shrink. A turnover tax asks the government mainly to verify revenue, which is harder to hide in a formal corporate tax system. Therefore, turnover taxation is a cruder but more enforceable instrument. The paper shows that the optimal tax base depends on a trade-off:

production distortion from a broad base   versus   evasion reduction from a broad base.

# 2 Incremental Contribution of the Paper

## 2.1 Contribution to optimal taxation in developing countries

The classic production-efficiency theorem says that tax systems should avoid taxing intermediate inputs, turnover, or trade when profit taxation and consumption taxation are available. This paper shows why that result can fail in practice: if governments cannot enforce a profit tax well, the formally efficient tax base may be fiscally inefficient.

The paper therefore connects optimal tax theory to the institutional reality of developing countries, where governments often rely on low-rate, broad-base taxes that economists usually view as distortive.

## 2.2 Contribution to empirical tax evasion measurement

The paper develops a quasi-experimental bunching method for estimating evasion responses. Its key innovation is to exploit a **nonstandard kink** where both the tax rate and the tax base change at the same point. The kink arises from the minimum tax rule:

$$T(y, \hat{c}) = \max\{\tau_p(y - \hat{c}), \tau_y y\},$$

where firms pay either a profit tax or a turnover tax, whichever liability is larger. The kink is located at

$$\hat{p} \equiv \frac{y - \hat{c}}{y} = \frac{\tau_y}{\tau_p}.$$

This allows the authors to infer how much evasion falls when firms move from profit-tax incentives to turnover-tax incentives.

## 2.3 Contribution to bunching methods

Standard bunching papers typically exploit kinks in marginal tax rates holding the tax base fixed. This paper studies a kink where the tax base changes from profit to turnover. Around this kink, the real production incentive changes only slightly because turnover tax rates are tiny, while the evasion incentive changes drastically because firms move from a profit tax base to a turnover tax base. Hence, large bunching is difficult to explain with real output responses and is instead strong evidence of evasion responses.

## 2.4 Contribution to policy design

The paper provides a framework for evaluating policies that are common in developing countries but often criticized as inefficient. It shows that minimum turnover taxes may be rational in environments where firms can evade profit taxes at low cost. The broader lesson is not that turnover taxes are always good, but that the optimal tax base depends on enforcement capacity.

# 3 Institutional Background: Pakistan's Minimum Tax Scheme

## 3.1 Corporate tax setting

Pakistan's corporation tax is a major federal revenue source. During the sample period, firms were subject to a standard profit tax rate and a minimum turnover tax. The policy required firms to calculate both liabilities and pay the larger one:

$$\text{Tax liability} = \max\{\text{profit tax liability}, \text{turnover tax liability}\}.$$

For most firms in 2006, 2007, and 2009, the profit tax rate was  $\tau_p = 35\%$ , and the turnover tax rate was  $\tau_y = 0.5\%$ . This generated a minimum-tax kink at

$$\frac{\tau_y}{\tau_p} = \frac{0.005}{0.35} = 1.43\%.$$

Firms reporting profit rates below 1.43 percent paid turnover tax; firms reporting profit rates above 1.43 percent paid profit tax.

### 3.2 Useful policy variation

The paper exploits several forms of variation:

- **Across firm types:** some low-rate firms faced a lower profit tax rate of 20 percent, moving the kink to  $0.005/0.20 = 2.5\%$ .
- **Across time:** the minimum tax was abolished in 2008, eliminating the kink.
- **Across time again:** the turnover tax rate doubled to 1 percent in 2010, shifting the high-rate-firm kink to  $0.01/0.35 = 2.86\%$ .

This variation allows the authors to check whether bunching moves exactly where the tax rule predicts.

## 4 Theoretical Framework

### 4.1 Firm problem

A firm produces output  $y$  at cost  $c(y)$ . It can misreport costs by declaring  $\hat{c} \neq c(y)$ , but doing so has a convex private and social cost  $g(\hat{c} - c(y))$ . The general tax liability is

$$T(y, \hat{c}) = \tau(y - \mu\hat{c}),$$

where  $\tau$  is the tax rate and  $\mu$  is the tax-base parameter. The parameter  $\mu$  ranges from

$$\mu = 0 \quad \text{turnover tax,} \quad \mu = 1 \quad \text{profit tax.}$$

The firm chooses output and reported cost to maximize after-tax profits:

$$\Pi(y, \hat{c}) = (1 - \tau)y - c(y) + \tau\mu\hat{c} - g(\hat{c} - c(y)).$$

### 4.2 Production efficiency versus evasion

The first-order conditions imply that the output choice is distorted by an effective marginal tax rate

$$\tau_E = \frac{\tau(1 - \mu)}{1 - \tau\mu}.$$

A pure profit tax,  $\mu = 1$ , has  $\tau_E = 0$ , so it preserves production efficiency. A turnover tax,  $\mu = 0$ , has  $\tau_E = \tau$ , so it distorts production.

But evasion increases with  $\tau\mu$ . A pure profit tax is easier to evade because costs are deductible and can be inflated. A turnover tax reduces the scope for cost evasion because costs do not enter the tax base.

### 4.3 Optimal tax-base rule

The partial-equilibrium optimality condition can be written as

$$\frac{\tau}{1-\tau} \frac{\partial \tau_E}{\partial \tau}(\mu) = G(\mu) \frac{\varepsilon_{\hat{c}-c}}{\varepsilon_y},$$

where  $G(\mu) = (\hat{c} - c)/\hat{\Pi}$  is evasion as a share of reported profits,  $\varepsilon_{\hat{c}-c}$  is the evasion elasticity, and  $\varepsilon_y$  is the real output elasticity.

**Interpretation:** the left-hand side is the production-efficiency cost of broadening the base, while the right-hand side is the revenue-efficiency gain from reducing evasion. In high-tax-capacity economies, the right-hand side is small, so the optimal base is close to profits. In low-tax-capacity economies, the right-hand side can be large, so a broader base can be optimal.

### 4.4 General equilibrium extension

The paper also extends the framework to intermediate goods. Turnover taxation creates cascading because the same value can be taxed multiple times as it passes through production stages. This adds a production-efficiency cost. At the same time, incidence effects can shift income between upstream and downstream sectors. The general equilibrium optimal rule therefore adjusts the production-efficiency term, but the core result remains: if evasion is important, some departure from production efficiency is optimal.

## 5 Empirical Methodology

### 5.1 Minimum tax kink

The minimum tax scheme creates a kink in reported profit rates:

$$\hat{p} = \frac{y - \hat{c}}{y} = \frac{\tau_y}{\tau_p}.$$

Firms below the kink face turnover taxation. Firms above the kink face profit taxation. Crossing the kink changes both the tax rate and the tax base.

### 5.2 Why bunching mainly captures evasion

At the kink, the real production incentive changes from a profit tax to a turnover tax. But the turnover tax rate is extremely small. In Pakistan,  $\tau_y = 0.5\%$ , so the real output incentive changes only slightly. By contrast, the evasion incentive changes dramatically: under profit taxation, inflating costs reduces tax liability; under turnover taxation, reported costs do not affect tax liability.

The model decomposes the profit-rate response as

$$\Delta \hat{p} \simeq \frac{\tau_y^2}{\tau_p} \varepsilon_y - \frac{d(\hat{c} - c)}{y}.$$

The first term is the real output response. Because  $\tau_y^2/\tau_p$  is tiny, the real response is small unless the output elasticity is implausibly large. The second term is the evasion response. Therefore, large bunching around the kink provides evidence of evasion.

### 5.3 Counterfactual density estimation

The authors estimate excess bunching by fitting a flexible polynomial to the observed density of reported profit rates, excluding the bunching region around the kink:

$$d_j = \sum_{i=0}^q \beta_i p_j^i + \sum_{i=p_L}^{p_U} \gamma_i \mathbf{1}[p_j = i] + \nu_j.$$

The counterfactual density is the fitted polynomial with the excluded-region dummies removed. Excess bunching is the difference between the observed density and this counterfactual density in the bunching window.

## 6 Data

### 6.1 Administrative corporate tax returns

The paper uses administrative data from Pakistan’s Federal Board of Revenue covering the universe of corporate income tax returns between 2006 and 2010. The data contain reported profits, turnover, tax liability, cost line items, and information needed to assign firms to the high-rate or low-rate schedule.

A major advantage is that electronic filing was mandatory for most firms after July 2007, reducing measurement error relative to many developing-country administrative data sources.

### 6.2 Sample construction

The analysis focuses on firms reporting both profits and turnover and enough information to determine their relevant tax schedule. The authors also apply consistency checks comparing reported and implied tax liabilities and taxpayer status. These checks are summarized in Appendix Tables A2 and A3.

## 7 Main Empirical Results

### 7.1 Bunching moves with the kink

The paper’s central nonparametric evidence is Figure 2. High-rate firms in 2006/7/9 bunch sharply at 1.43 percent. Low-rate firms bunch at 2.5 percent. When the minimum tax scheme is abolished in 2008, the bunch disappears. When the turnover tax doubles in 2010, the bunch shifts to 2.86 percent. This pattern strongly supports the interpretation that firms respond to the minimum tax kink.

### 7.2 Evasion responses are large

The baseline estimates imply that for high-rate firms in 2006/7/9, the bunching estimate is  $b = 4.47$ , corresponding to a profit-rate response of about 1 percentage point. Without evasion, this bunching would require an output elasticity of about 134, which is implausible. With evasion, it

corresponds to an evasion reduction of about 67 percent of taxable profits when moving from profit to turnover taxation. The result remains very similar even when assuming extremely high real output elasticities.

### 7.3 Lazy reporting is not the main explanation

An alternative explanation is that firms in the turnover-tax regime may lazily omit cost categories because those costs no longer affect tax liability. The paper tests this using the number of cost categories reported on the tax return. Figure 4 shows that the fraction of cost categories used varies smoothly across the kink and does not fall discretely under the turnover tax. This supports the interpretation that bunching reflects deliberate evasion responses rather than lazy reporting errors.

### 7.4 Policy calibration

The calibrated model shows that switching from profit taxation to turnover taxation can raise revenue by 74 percent while holding aggregate after-tax profits constant. The pure turnover tax realizes almost all the gains from the fully optimal tax system in the calibration. This is because the compliance gains are large relative to the production-efficiency cost.

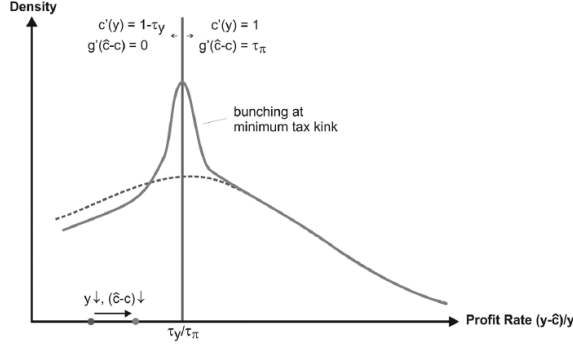
## 8 Figures and Tables

### 8.1 Figure 1: Bunching Methodology Using Minimum Tax Schemes; Table 1: Descriptive Statistics

**Read:** Figure 1 provides the conceptual bunching logic. Firms below the minimum tax cutoff face turnover taxation, reduce evasion, and move their reported profit rate toward the kink. Table 1 shows the institutional tax rates and the importance of the turnover-tax regime in Pakistan.

**Economic message:** The minimum tax is not a small institutional detail. In the sample years, roughly half or more of firms are in the turnover-tax regime, and turnover tax accounts for a large share of corporate tax revenue.





(a) Figure 1: Bunching methodology

|                                       | 2006            | 2007            | 2008            | 2009            | 2010            |
|---------------------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| A. Tax Variables                      |                 |                 |                 |                 |                 |
| Profit tax rate (high)                | .35             | .35             | .35             | .35             | .35             |
| Profit tax rate (low)                 | .2              | .2              | .2              | .2              | .25             |
| Turnover tax rate                     | .005            | .005            | NA              | .005            | .01             |
| Share of firms in turnover tax regime | .498            | .561            | NA              | .532            | .621            |
| Share of revenue from turnover tax    | .655            | .713            | NA              | .581            | .751            |
| B. Firm Characteristics (Means)       |                 |                 |                 |                 |                 |
| Profits (million PKR)                 | 1.1<br>(.2)     | .1<br>(.2)      | -.4<br>(.1)     | .3<br>(.1)      | 1.1<br>(.2)     |
| Turnover (million PKR)                | 290.9<br>(24.2) | 369.3<br>(83.9) | 246.1<br>(57.5) | 258.1<br>(24.3) | 275.6<br>(23.7) |
| Profit rate (profits/turnover)        | 2.08<br>(.1)    | 2.06<br>(.06)   | 1.95<br>(.04)   | 2.2<br>(.05)    | 2.46<br>(.05)   |
| Salary/turnover                       | .182<br>(.005)  | .200<br>(.005)  | .210<br>(.004)  | .217<br>(.004)  | .235<br>(.005)  |
| Interests/turnover                    | .019<br>(.001)  | .018<br>(.002)  | .018<br>(.002)  | .016<br>(.001)  | .018<br>(.001)  |
| Observations                          | 8,604           | 14,587          | 20,485          | 19,944          | 19,909          |

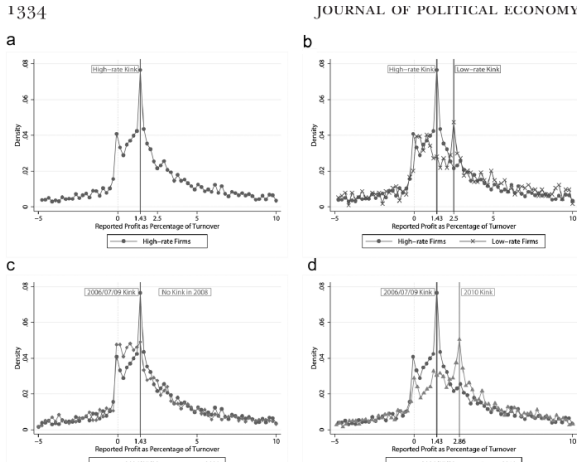
NOTE.—The table presents descriptive statistics, focusing on tax variables (panel A) and firm characteristics (panel B). Rows 1–3 of panel A are based on Pakistan's corporate tax schedule. The remaining rows are based on administrative tax return data from the universe of tax-registered firms in Pakistan. All statistics are based on the raw data, excluding each year the top and bottom 5 percent tails in terms of profits. Standard errors are shown in parentheses. The low tax rate applies to firms that registered after June 2005, have no more than 250 employees, have annual sales of not more than Rs. 250 million, and have paid-up capital of not more than Rs. 25 million (all monetary figures are given in Pakistani rupees (PKR)).

(b) Table 1: Descriptive statistics

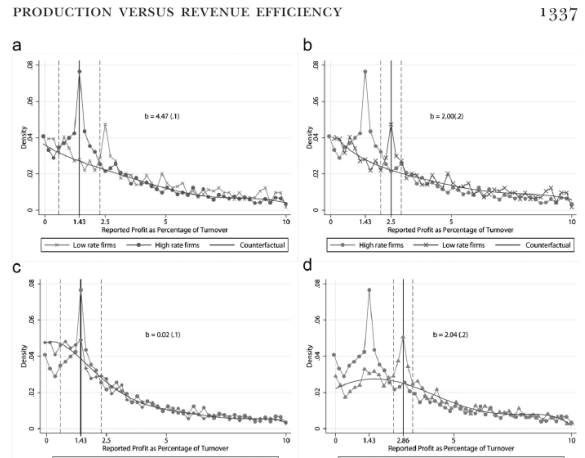
## 8.2 Figure 2: Bunching Evidence; Figure 3: Bunching Estimation

**Read:** Figure 2 shows raw bunching at the kink and how the bunch moves or disappears when the kink changes. Figure 3 estimates counterfactual distributions and excess mass.

**Economic message:** The bunching evidence is compelling because it is not fixed at one arbitrary profit rate. It moves with the tax schedule, appears when the minimum tax exists, and disappears when the minimum tax is temporarily abolished.



(a) Figure 2: Bunching evidence



(b) Figure 3: Bunching estimation

## 8.3 Table 2: Estimating Evasion Responses; Figure 4: Usage of Cost Categories Varies Smoothly across the Kink

**Read:** Table 2 converts bunching into real-output and evasion-response estimates. Figure 4 checks whether firms report fewer cost categories under turnover taxation.

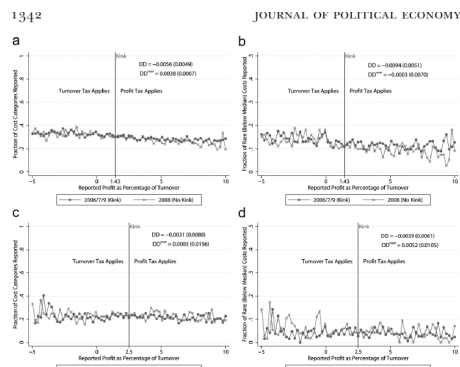
**Economic message:** Real output responses alone would require implausibly large elasticities, so the observed bunching is mostly evasion. The smooth cost-category pattern rejects the simple lazy-reporting explanation.

TABLE 2  
ESTIMATING EVASION RESPONSES

|                           | OBSERVED RESPONSES      |                                             | MODEL WITHOUT EVASION<br>Output Elasticity ( $\epsilon_y$ ) | MODEL WITH EVASION<br>Evasion Rate Response |                          |                         |                         |
|---------------------------|-------------------------|---------------------------------------------|-------------------------------------------------------------|---------------------------------------------|--------------------------|-------------------------|-------------------------|
|                           | Bunching ( $b$ )<br>(1) | Profit Rate Response ( $\Delta\pi$ )<br>(2) |                                                             | $\epsilon_y = 0$<br>(4)                     | $\epsilon_y = .5$<br>(5) | $\epsilon_y = 1$<br>(6) | $\epsilon_y = 5$<br>(7) |
| High-rate firms, 2006/7/9 | 4.47<br>(.1)            | 1.0<br>(.03)                                | 134.2<br>(4.0)                                              | 67.1<br>(2.0)                               | 66.9<br>(2.0)            | 66.6<br>(2.0)           | 64.6<br>(2.0)           |
| Low-rate firms, 2006/7/9  | 2.00<br>(.2)            | .4<br>(.04)                                 | 34.3<br>(3.1)                                               | 17.1<br>(1.6)                               | 16.9<br>(1.6)            | 16.6<br>(1.6)           | 14.6<br>(1.6)           |
| High-rate firms, 2010     | 2.04<br>(.1)            | .4<br>(.03)                                 | 14.6<br>(1.1)                                               | 14.6<br>(1.1)                               | 14.1<br>(1.1)            | 13.6<br>(1.1)           | 9.6<br>(1.1)            |

NOTE.—This table presents bunching, elasticity, and evasion estimates for the subsamples considered in panels a, b, and d of fig. 3. Column 1 reproduces the bunching estimate  $b$ , based on estimating eq. (14). Bunching  $b$  is the excess mass in the excluded range around the kink, in proportion to the average counterfactual density in the excluded range. Column 2 presents an estimate of the profit rate response (in percent) associated with  $b$ , based on the relationship  $\Delta\pi = B/b \cdot (r_y/r_x) \approx b \times \text{bin width}$ . Column 3 presents estimates of the real output elasticity  $\epsilon_y$  for the model without evasion. This model is based on the assumption that bunching is purely due to a real output response. The elasticity  $\epsilon_y$  is estimated using the relationship  $\Delta\pi = [c/y - c'(y)]dy/y \approx (r_y/r_x)\epsilon_y$ . Columns 4–7 present estimates of the evasion response as a percentage of taxable profits (evasion rate responses) for the model with evasion. This model allows for bunching to be driven by both evasion and real output response. The evasion response estimates are based on  $\Delta\pi = [c/y - c'(y)]dy/y - [d(\bar{c} - c)/y] \approx (r_y/r_x)\epsilon_y - (r_y/r_x)[d(\bar{c} - c)/y - \bar{c}]$ , assuming different real output elasticities  $\epsilon_y \in \{0, 0.5, 1, 5\}$ . Bootstrapped standard errors are shown in parentheses.

(a) Table 2: Evasion responses



(b) Figure 4: Cost categories

## 8.4 Table 3: Numerical Tax Policy Analysis; Table A1: Comparison of Missing and Nonmissing Observations

**Read:** Table 3 reports the numerical tax-policy simulation. Table A1 compares missing and nonmissing observations.

**Economic message:** The calibrated revenue gains from a turnover tax are large, and the appendix table helps clarify the sample coverage and selection issues.

TABLE 3  
NUMERICAL TAX POLICY ANALYSIS

| OUTPUT<br>ELASTICITY ( $\epsilon_y$ ) | A. PURE TURNOVER TAX       |                              |                               | B. OPTIMAL TAX             |                              |                               |
|---------------------------------------|----------------------------|------------------------------|-------------------------------|----------------------------|------------------------------|-------------------------------|
|                                       | Revenue<br>Gain (%)<br>(1) | Tax<br>Base ( $\mu$ )<br>(2) | Tax<br>Rate ( $\tau$ )<br>(3) | Revenue<br>Gain (%)<br>(4) | Tax<br>Base ( $\mu$ )<br>(5) | Tax<br>Rate ( $\tau$ )<br>(6) |
| .5                                    | 74                         | 0                            | .005                          | 76                         | .522                         | .009                          |
| 1                                     | 73                         | 0                            | .005                          | 76                         | .706                         | .015                          |
| 5                                     | 70                         | 0                            | .005                          | 75                         | .889                         | .037                          |
| 10                                    | 66                         | 0                            | .005                          | 75                         | .944                         | .067                          |
| 30                                    | 62                         | 0                            | .005                          | 77                         | .986                         | .170                          |

NOTE.—This table presents a numerical evaluation of the tax policy changes discussed in Sec. VI. For an assumed output elasticity, we calibrate firm-specific production parameters and cost evasion parameters to replicate the empirical distributions of turnover and costs and to match our evasion response estimates. All details of the calibration are in Sec. C of the Appendix. In panel A, we simulate a switch from a pure profit tax ( $\tau = 0.35$ ;  $\mu = 1$ ) to a pure turnover tax, adjusting the tax rate so that aggregate (after-tax) profits remain the same. In panel B, we simulate a switch from the same pure profit tax ( $\tau = 0.35$ ;  $\mu = 1$ ) to the optimal tax policy, maximizing tax revenues without reducing aggregate profits. Columns 1 and 4 show the implied tax revenue gain for the pure turnover tax and the optimal tax, respectively. Columns 2 and 5 show the corresponding tax base parameters. Columns 3 and 6 show the corresponding tax rates. The results illustrate that turnover taxation dominates profit taxation and realizes almost all potential welfare gains from setting the tax base and rate optimally.

(a) Table 3: Numerical tax policy analysis

TABLE A1  
COMPARISON OF MISSING AND NONMISSING OBSERVATIONS

|                                         | Observations<br>(1) | Median<br>(2) | Mean<br>(3) | Standard<br>Deviation<br>(4) |
|-----------------------------------------|---------------------|---------------|-------------|------------------------------|
| A. Firms Reporting Profits and Turnover |                     |               |             |                              |
| Profits                                 | 15,681              | .1            | −.7         | 12.3                         |
| Turnover                                | 15,681              | 18.7          | 181.0       | 703.5                        |
| Salary                                  | 6,714               | 6.3           | 25.9        | 76.6                         |
| Interest                                | 8,361               | .5            | 10.7        | 49.9                         |
| Share of low-rate firms                 | 15,681              |               | .20         |                              |
| B. Firms Reporting Profits Only         |                     |               |             |                              |
| Profits                                 | 11,754              | .0            | −.8         | 12.5                         |
| Salary                                  | 2,800               | 9.1           | 35.9        | 94.5                         |
| Interest                                | 4,185               | .2            | 13.1        | 74.7                         |
| Share of low-rate firms                 | 10,467              |               | .16         |                              |
| C. Firms Reporting Turnover Only        |                     |               |             |                              |
| Turnover                                | 8,546               | 9.2           | 454.5       | 7,401.3                      |
| Salary                                  | 3,073               | 5.0           | 37.9        | 272.0                        |
| Interest                                | 3,763               | .7            | 40.2        | 260.1                        |
| Share of low-rate firms                 | 8,546               |               | .27         |                              |

NOTE.—The table compares different samples of firms depending on whether or not they report profits and turnover. This is based on the final data (after consistency checks are applied) for all years pooled, excluding each year in the top and bottom 5 percent tails in terms of profits. Panel A considers firms that report both profits and turnover. Panel B considers firms that report profits only. Panel C considers firms that report turnover only. Columns 1–4 report the number of observations, median, mean, and standard deviation for different observable characteristics (turnover, profits, salary payments, interest payments, share of small firms). All statistics are in millions of Pakistani rupees.

(b) Table A1: Missing versus nonmissing observations

## 8.5 Table A2: Data Cleaning Steps - Sample Definition; Table A3: Data Cleaning Steps - Sample Size

**Read:** Table A2 explains the internal consistency checks. Table A3 reports how the sample changes through each cleaning step.

**Economic message:** These tables are important because the paper relies on administrative tax returns, and the validity of the bunching analysis depends on reliable assignment of firms to the correct tax regime and kink.

| TABLE A2<br>DATA CLEANING STEPS: SAMPLE DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sample                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Definition                                                                                                                                                                                                                                                                                  |
| Firms reporting profits and turnover                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Firms reporting profits $\Pi$ , turnover $y$ , and incorporation date $D$ . Based on $\Pi$ and $y$ , derive implied tax liabilities $\hat{T}$ , $\hat{T}_H$ , and $\hat{T}_L$ (high and low profit rate).                                                                                   |
| Consistency check I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Drop firm if reported and implied tax liability are inconsistent, i.e., $T^y \neq \hat{T}$ or $T^H \neq \hat{T}_H$ and $T^L \neq \hat{T}_L$ . If $T^H = \hat{T}_H$ or $T^L = \hat{T}_L$ , assign $\{H, L\}$ . If $T^H$ is missing, assign $\{H, L\}$ based on $y$ , $D$ , and capital $K$ . |
| Consistency check II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Drop firm if reported and implied taxpayer status are inconsistent, i.e., if $T^y > \hat{T}^H$ and $\hat{T} < \hat{T}^H$ ; $T^y < \hat{T}^H$ and $\hat{T} > \hat{T}^H$ ; $\hat{T} > \hat{T}^H$ and $T^y$ are missing; and $\hat{T} < \hat{T}^H$ and $T^H$ are missing.                      |
| NOTE.—This table explains the consistency checks applied to the data. For all consistency checks, a tolerance threshold of 5 percent is used. Capital $K$ is equity plus retained earnings. Note that the implied turnover tax liability used for consistency check I is gross implied turnover tax liability minus net deductions (which are deducted from the tax liability before the taxpayer status—turnover or profit taxpayer—is determined). For the same reason, the profits to turnover ratio used for consistency check II and for the bunching graphs is (profits minus net reductions)/turnover for firms that report positive net reductions. |                                                                                                                                                                                                                                                                                             |

(a) Table A2: Sample definition

tableA3\_sample\_size.png

(b) Table A3: Sample size

## 8.6 Table A4: Robustness of Bunching Estimates

**Read:** Table A4 shows how bunching estimates change when the polynomial order and the excluded range around the kink are varied.

**Economic message:** The bunching estimates are not purely mechanical; they are sensitive to specification choices, especially for high-rate firms in 2006/7/9. This is common in bunching designs and should be remembered when interpreting magnitudes, even though the qualitative evidence of bunching is very strong.

TABLE A4  
ROBUSTNESS OF BUNCHING ESTIMATES

|                           | A. VARYING THE ORDER OF POLYNOMIAL     |              |              |              |              |
|---------------------------|----------------------------------------|--------------|--------------|--------------|--------------|
|                           | 3                                      | 4            | 5            | 6            | 7            |
| High-rate firms, 2006/7/9 | 4.27<br>(.1)                           | 3.92<br>(.1) | 4.47<br>(.1) | 6.13<br>(.1) | 5.59<br>(.1) |
| Low-rate firms, 2006/7/9  | 1.93<br>(.2)                           | 2.04<br>(.2) | 2.00<br>(.2) | 2.47<br>(.2) | 2.50<br>(.2) |
| High-rate firms, 2010     | 2.53<br>(.2)                           | 2.23<br>(.2) | 2.04<br>(.2) | 1.48<br>(.1) | 1.41<br>(.1) |
|                           | B. VARYING THE NUMBER OF EXCLUDED BINS |              |              |              |              |
|                           | 1                                      | 2            | 3            | 4            | 5            |
| High-rate firms, 2006/7/9 | 1.83<br>(.1)                           | 2.58<br>(.1) | 3.68<br>(.1) | 4.47<br>(.1) | 2.22<br>(.1) |
| Low-rate firms, 2006/7/9  | 1.70<br>(.1)                           | 2.00<br>(.2) | 2.01<br>(.2) | 1.48<br>(.3) | 1.45<br>(.3) |
| High-rate firms, 2010     | 1.81<br>(.1)                           | 2.04<br>(.2) | 2.55<br>(.2) | 2.41<br>(.2) | 2.31<br>(.3) |

NOTE.—The table presents estimates of the excess mass  $\hat{b}$ , for different specifications of the estimating eq. (14), for the subsamples considered in table 2. Bunching  $\hat{b}$  is the excess mass in the excluded range around the kink, in proportion to the average counterfactual density in the excluded range. Panel A presents estimates for different choices of the order of polynomial  $q \in \{3, 4, 5, 6, 7\}$ , for the excluded range chosen as in table 2 (four bins on either side of the kink for high-rate firms in 2006/7/9, two bins otherwise). Panel B presents estimates for different choices of the excluded range (one to five bins on either side of the kink) for the order of polynomial chosen as in table 2 ( $q = 5$ ). Bootstrapped standard errors are shown in parentheses.

Table A4: Robustness of bunching estimates

## 9 Conclusion

### 9.1 Core conclusion

The paper shows that the standard production-efficiency prescription can fail when tax enforcement is weak. If profit taxes are easy to evade, a turnover tax can raise revenue and welfare despite distorting production. In Pakistan, minimum-tax kinks reveal large evasion responses, implying that turnover taxation can sharply reduce evasion.

### 9.2 Why the paper matters

The paper is important because it turns a common developing-country policy, often dismissed as inefficient, into a source of identification and policy learning. It shows that the correct benchmark is not an idealized efficient tax system with perfect enforcement, but a constrained system in which evasion and enforcement are first-order.

### 9.3 Limitations

The estimates capture intensive-margin responses among corporate tax filers. They do not directly estimate incorporation responses, informality responses, or full general-equilibrium cascading effects. The calibration is partial equilibrium, although the paper discusses how general equilibrium forces could matter.

### 9.4 Takeaway

The key takeaway is that production-inefficient instruments may be revenue-efficient when state capacity is limited. For developing countries, tax design must jointly consider efficiency, evasion, enforcement, and administrative simplicity.